# A Machine Learning Approach to Volatility Forecasting

*Replication and critical discussion of Christensen, Siggaard & Veliyev (2023, Journal of Financial Econometrics 21(5)) — AAPL, AMZN, JPM 5-minute data, 2016–2024.*

## 1. The problem and the original paper

Realised variance (RV), the sum of squared intraday returns, is the ex-post measure of a day's return variation (Andersen & Bollerslev 1998) and the input to risk management, option pricing and allocation; unlike returns it is strongly predictable because volatility is persistent. The field benchmark is the Heterogeneous Autoregressive model (HAR; Corsi 2009): a linear regression of RV on its daily, weekly (5-day) and monthly (22-day) lagged averages — four parameters that approximate the long memory of volatility so well that decades of richer models struggle to beat it (Hansen & Lunde 2005).

Christensen, Siggaard & Veliyev (2023, "CSV") ask whether machine learning (ML) beats HAR, and where any gain comes from. They run a 22-model horse-race — the HAR family (HAR, LogHAR, LevHAR, SHAR, HARQ), regularised regression (Ridge, Lasso, Elastic Net, adaptive- and post-Lasso), tree ensembles (bagging, random forest, gradient boosting) and feed-forward neural nets (four geometric-pyramid architectures, each a single net $NN^1$ and a 10-of-100 ensemble $NN^{10}$) — over two information sets: M_HAR (three RV lags) and M_ALL (plus nine macro/firm predictors). On 29 Dow Jones stocks, 2001–2017, they conclude ML beats HAR; the gain is small and class-specific on M_HAR but clear on M_ALL (Elastic Net ≈–8%, random forest ≈–10%, best NN ≈–11% at h=1), grows with horizon, and is led by random forests at one month (≈40% MSE reduction). Significance uses Diebold–Mariano (DM; 1995) and the Hansen–Lunde–Nason Model Confidence Set (MCS; 2011); interpretability uses Accumulated Local Effects (Apley & Zhu 2020).

We replicate the full methodology on the three stocks with available 5-minute data over 2016–2024 — seven years later than CSV, spanning the COVID shock and 2022 regime. AAPL and JPM are in CSV's cross-section (a check); AMZN joined the DJIA only in 2020, giving out-of-sample-stock evidence. We target a sound, transparent implementation and a critical comparison, not exact numbers.

## 2. Implementation and replication

**Data.** From 1-minute bars we keep the 09:30–16:00 session, apply a Barndorff-Nielsen et al. (2009)-style outlier filter, resample to 5-minute returns within each day (no overnight returns), and compute RV, signed semivariances and realised quarticity. The split is 70/10/20 chronological, as CSV. The daily lag $RV_{t-1}$ predicts $RV_t$ (a true one-step forecast).

**Models.** All 22 CSV models sit behind one interface. HAR is closed-form OLS; LogHAR carries the Jensen back-transform; HARQ uses the Bollerslev–Patton–Quaedvlieg (2016) insanity filter (clip to the in-sample range). Regularised models use a 100-point log-λ grid (CSV 1,000); the resolution is immaterial, but the selected λ often sits at a grid bound (Ridge at the upper bound on all three stocks; see selected_lambda.csv), so the grid range, not its density, binds. Bagging/RF use Breiman–Cutler defaults; gradient boosting is validation-tuned. Following CSV (p.1691) any negative variance forecast is replaced by the in-sample minimum RV, applied uniformly.

**Honest deviations.** The NNs run on scikit-learn (PyTorch unavailable): ReLU not leaky-ReLU and no dropout, so early stopping is the regulariser in dropout's place. For tractability the regularised, tree and NN models are fixed-window; CSV roll the regularised models and trees and fix only the NNs — our fixed-window trees are the main deviation, exploited in §4. Implied volatility is proxied by VIX (CSV use licensed OptionMetrics). **Evaluation:** MSE (primary, as CSV) plus QLIKE (Patton 2011) as a noise-robust complement; DM with Newey–West HAC and the Harvey–Leybourne–Newbold correction; the HLN MCS (10,000 moving-block bootstraps) at $\alpha \in \{0.25, 0.10, 0.05\}$; ALE for variable importance. Thirty-seven unit tests cover the critical path.

## 3. Results

We select three results as most important — the M_ALL comparison at h=1, its behaviour across horizons, and the significance picture — because these are CSV's three headline claims.

| h | Stock | LogHAR | EN | LA | RF | BG | $NN2^{10}$ | $NN3^{10}$ |
|---|---|---|---|---|---|---|---|---|
| 1 | AAPL | 0.944 | 0.953 | 0.941 | 1.069 | 1.167 | 0.891 | 0.938 |
| 1 | AMZN | 1.000 | 0.970 | 0.881 | 0.963 | 1.140 | 0.879 | 0.861 |
| 1 | JPM | 0.924 | 0.953 | 0.951 | 0.952 | 1.054 | 0.912 | 0.950 |
| 22 | AAPL | 0.666 | 1.031 | 1.021 | 3.038 | 7.314 | 1.009 | 0.926 |
| 22 | AMZN | 1.214 | 1.085 | 0.901 | 3.885 | 6.719 | 1.129 | 1.245 |
| 22 | JPM | 0.630 | 1.012 | 0.976 | 5.274 | 13.835 | 0.860 | 0.716 |

*Table 1. Out-of-sample MSE relative to HAR, M_ALL (ratio < 1 beats HAR). Full 22-model tables, DM stars and MCS sets in the Appendix.*

**At h=1 on M_ALL** the best model beats HAR by ≈9–14% on every stock (AAPL 10.9%, AMZN 13.9%, JPM 8.8%) — NN ensembles lead (0.86–0.91), Lasso close (0.88–0.95) — reproducing CSV's magnitude (≈11%) and ordering (NNs best, regularisation helping, bagging weakest). On the apples-to-apples M_HAR set the gain shrinks to a few percent and only the NN ensembles beat HAR, with trees losing — again matching CSV.

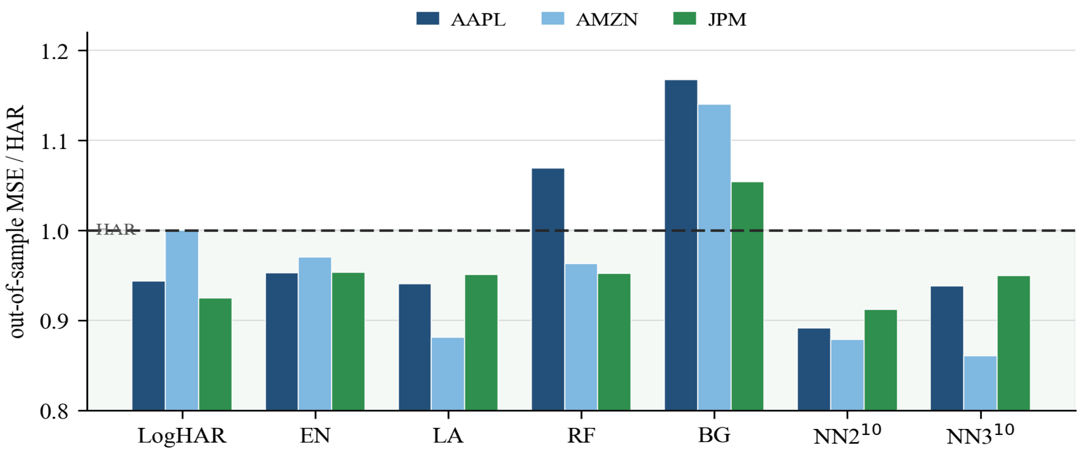


*Figure 1. M_ALL MSE ratios vs HAR at h=1; bars below the dashed HAR line beat HAR.*

**Across horizons** the tree picture diverges sharply and is our most informative result. CSV find ML's edge grows with horizon, RF leading at h=22; we find the opposite for fixed-window trees — at h=22 bagging is 7–14× HAR and RF 3–5× — while LogHAR dominates the HAR family (0.63–0.67 on AAPL/JPM) and the NN ensembles stay competitive. **Significance is thin:** at h=1 the MCS at 90% retains 20–21 of 22 models on every stock and never eliminates HAR; one-sided DM finds only 0–3 models per stock beat HAR at 5%, and under a Holm/BH multiple-testing correction almost nothing survives. Under QLIKE the ML advantage largely disappears.

## 4. Critical comparison and discussion

The replication matches CSV where the data allow (LogHAR strongest in the HAR family, NN ensembles best at h=1, regularisation helping on M_ALL), so the substantive contribution is the critique. Five points, each tied to a financial, econometric or ML reason; full development with page/table references in the Appendix.

**(i) The "machine learning" headline is mostly information set and regularisation, not nonlinearity [ML].** CSV's own tables separate the effects their title bundles. On M_HAR (Table 2, h=1) only the NN ensembles beat HAR, by ≈5%; trees lose (bagging +15%) and regularisation ties (EN 0.999). The gain appears only on M_ALL, where plain OLS on the wider set (HAR-X) already earns 3.4%, Elastic Net 8.4% and the best NN 11.5% — regularisation does roughly half the work, nonlinearity the rest. Our path decomposition reproduces the split (regularisation +3–5pp, deep-NN +4–18pp, trees flat-to-negative). "Regularisation on a richer information set" would describe the result more honestly than "machine learning".

**(ii) CSV's long-horizon tree result depends on the daily re-fit, not the model class [econometric/financial].** CSV roll bagging/RF daily but fix the NNs once (p.1693, conceded "strictly disadvantageous to the NNs"), so "RF best at h=22" (Table 7, ≈40% reduction) confounds *trees* with *re-estimated daily*. We isolate the two: with all ML fixed-window the trees collapse at h=22; refitting RF daily — CSV's scheme — roughly halves the AAPL ratio (3.04→1.55), pinning the daily re-fit, not "trees capture long memory", as the load-bearing assumption (it does not fully reach CSV's RF<HAR, evidence the result is also sample-conditional). A model needing constant re-estimation to survive is the one that fails at a regime change, when forecasts matter most.

**(iii) The significance machinery does not support the prose [econometric].** CSV's claim is a between-means statement (lower average MSE via DM). The MCS — the test of *distinguishable* accuracy — keeps HAR in the best set: their Figure 4 retains LogHAR at a high rate, and ours retains 20–21 of 22 models at α=0.10 with HAR never eliminated. DM finds 0–3 models per stock beat HAR at 5%, and a multiple-testing correction (CSV apply none, despite ≈136 pairwise tests per stock) removes essentially every h=1/h=5 star. ML lowers average MSE a few percent but is not statistically distinguishable from HAR at the daily horizon.

**(iv) The edge is loss- and regime-conditional [financial].** CSV evaluate on MSE only; under QLIKE (Patton's noise-robust loss) the advantage largely vanishes — HAR-X, Ridge, Elastic Net and the NNs are all QLIKE-worse than HAR on most stocks at h=1. MSE rewards fit on a few high-variance days; QLIKE weights the many ordinary days, where HAR is hard to beat. And on a 2016–2019-train / 2020–2024-test split (spanning the COVID crash and 2022) the edge collapses from ≈9–14% to ≈0–2% — no model beats HAR on JPM. Since proportional accuracy and turbulent regimes are what risk management cares about, the practical value of the edge is smaller than an MSE-only, calm-period comparison implies.

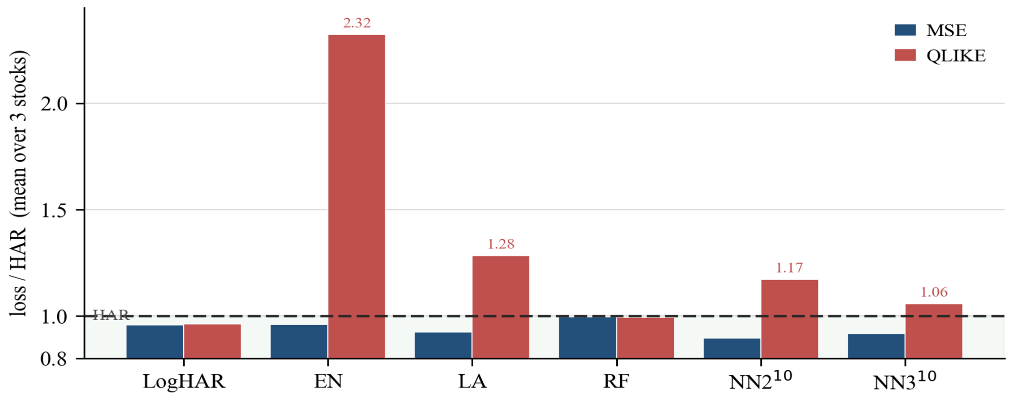


*Figure 2. Models that beat HAR under MSE (left bars < 1) are mostly worse under the noise-robust QLIKE loss (right bars > 1); only LogHAR is robust to both. h=1, M_ALL, mean over three stocks.*

**(v) Long-memory interpretation and licensed-data dependence [econometric/reproducibility].** CSV credit "better approximating long memory", but their evidence is *in-sample fitted* ACF (Figure 8), which cannot distinguish captured long memory from over-smoothing toward the unconditional mean — our *out-of-sample* ACF shows ML forecasts more persistent than realised RV, the over-smoothing signature. And the M_ALL gain leans on per-stock OptionMetrics implied volatility, the strongest extra predictor (Table A.5, t=6.4) yet licensed, unreproducible, and redundant once proxied by VIX (insignificant alongside IV, t=0.09). Where we match CSV the agreement is reassuring; where we differ, the difference is the contribution — enabled by a regime-spanning sample CSV could not observe.

**Further critiques (full development in `CRITIQUE.md`). (vi)** The comparison runs on a noisy proxy: RV estimates integrated variance with error and HARQ exists to correct it (BPQ 2016), yet the ML methods receive the same noisy regressors uncorrected — switching to the noise-robust realised kernel moves JPM h=1 MSE by ≈27%, larger than the entire ML-vs-HAR gap, so part of the "gain" is estimator noise, not model skill [econometric]. **(vii)** "Minimal tuning" is asymmetric: CSV search a 1,000-point $\lambda$ grid and keep the best 10 of 100 NN seeds but leave RF at 2004 defaults — the methods that win are the most heavily searched [ML]. **(viii)** The 22-model space is narrow: every model's out-of-sample errors correlate with HAR at $r \geq 0.79$ (≈0.88–0.99 for the models that beat HAR), so each winner is HAR plus a small orthogonal perturbation and its gain lives in under ~20% of error variance — which is also why combinations add little [ML]. **(ix)** "The basic HAR is superior to its offspring" (CSV p.1695) is split-dependent: their own Appendix A.1 flips it under a shorter window, so the headline ranking is conditional on the train/test design rather than structural [econometric].

## References

Andersen & Bollerslev (1998, *Int. Econ. Rev.*); Apley & Zhu (2020, *JRSS-B*); Barndorff-Nielsen & Shephard (2002, *JRSS-B*); Barndorff-Nielsen, Hansen, Lunde & Shephard (2008, *Econometrica*); Bollerslev, Patton & Quaedvlieg (2016, *J. Econometrics*); Christensen, Siggaard & Veliyev (2023, *J. Financial Econometrics* 21(5), 1680–1727); Corsi (2009, *J. Financial Econometrics*); Diebold & Mariano (1995, *JBES*); Hansen & Lunde (2005, *J. Applied Econometrics*); Hansen, Lunde & Nason (2011, *Econometrica*); Patton (2011, *J. Econometrics*); Patton & Sheppard (2015, *Rev. Econ. Stat.*); Zou (2006, *JASA*).

## Appendix (≤3 pages — full content in APPENDIX.md)

A1 full 22-model MSE tables (M_ALL h=1 & h=22); A2 the apples-to-apples M_HAR table; A3 critique evidence (path decomposition, fixed-vs-rolling tree experiment, MCS/DM significance, QLIKE reversal, realised-kernel, error correlation); A4 ALE variable importance; A5 code excerpts; A6 data construction and disclosed deviations. Set the wide A1 tables landscape (as CSV do). Machine-readable tables in `outputs/`; full critique in `CRITIQUE.md`.